

1. Course number and name

CHE 426. Biopharmaceutical Process Development and Manufacturing

2. Credits, contact hours and categorization

Credit hours: 3 or 4. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Nandkishor Nere and Meenesh Singh

4. Text book, title, author, publisher, year

- a) D. J. am Ende and M. T. am Ende (Editors), "Chemical Engineering in the Pharmaceutical Industry: Active Pharmaceutical Ingredients", 2nd Edition, Hoboken, NJ, Wiley, 2019.
- b) D. J. am Ende and M. T. am Ende (Editors), "Chemical Engineering in the Pharmaceutical Industry: Drug Product Design, Development, and Modeling", 2nd Edition, Hoboken, NJ, Wiley, 2019.
- c) F. Jameel, S. Hershenson, M. A. Khan, and S. Martin-Moe (Editors), "AAPS Advances in the Pharmaceutical Sciences Series: Quality by Design for Biopharmaceutical Drug Product Development (Vol 18)", Springer, 2015.
- d) F. Jameel, and S. Hershenson (Editors), "Formulation and Process Development Strategies for Manufacturing Biopharmaceuticals", Hoboken, NJ, Wiley, 2010.

Other supplemental materials

- a) N. K. Pandit and R. P. Soltis, "Introduction to the Pharmaceutical Sciences: An Integrated Approach", 2nd Edition, Lippincott Williams & Wilkins, 2012.
- b) N. W. Warne and H-C. Mahler (Editors), "AAPS Advances in the Pharmaceutical Sciences Series: Challenges in Protein Product (Vol 38)", Springer, 2015.

5. Specific course information

- a. Brief description of the content of the course (catalog description)
Introduction to Biopharmaceuticals Process Development and Manufacturing. Drug Substance (Active Pharmaceutical Ingredients) Development. Drug Product Development. Enabling Technologies in Biopharmaceutical Industry. Biopharmaceutical Manufacturing.
- b. Prerequisites or co-requisites
Consent of the instructor.
- c. Indicate whether a required, elective, or selected elective.
Elective.

6. Specific goals for the course

- a. Specific outcomes of instruction.
To be able to understand manufacturing processes for both the small molecules and biologics. Learn how to apply fundamentals of chemical engineering unit operations and

processes to design and scale-up processes to make both the drug substances and drug products. Use integrated engineering principles to solve practical problems of process development in biopharmaceutical industry. Work in collaboration with peers to solve problems assigned through the team project and communicate through interactive project presentations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.				X
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			X	
(3)	an ability to communicate effectively with a range of audiences.				X
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	X			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.				X
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.			X	
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		X		

7. Brief list of topics to be covered

The course is aimed at senior undergraduate and graduate students spanning multiple disciplines and is designed to introduce the practical world of process development and manufacturing of pharmaceutically active molecules and drug products.

- Introduction to drug discovery, drug delivery and development.
- Bio/Analytical Chemistry Tools and Technologies
- Active pharmaceutical ingredients (API)-Solid Form Screening and Selection
- Essentials of API Process Development
 - Fundamentals of chemical reaction engineering (CRE)
 - Applications of CRE in pharmaceutical industry
 - Crystallization process development
 - Extraction, Distillation, Filtration, Washing and Drying
- Biopharmaceutical Process Scale-up
- Essentials of Biologics Process Development
 - Upstream (bioreactor) process development
 - Downstream (purification) process development
- Essentials of Small Molecule Drug Product Formulation and Process Development
 - Formulation Screening and Selection

- Blending, Roller compaction, Wet Granulation, Film coating
 - Spray Drying, Hot Melt Extrusion, Tableting and Encapsulation
- Essentials of Biologics Drug Product Development
 - Formulation, overall process and lyophilization
 - Freeze-Thawing, Pooling, Reconstitution and Filling
- Advanced Topics
 - Applications of Predictive Modeling
 - Continuous Processing
 - Emerging Technologies-Electrochemistry, Photochemistry, Bio-catalysis, Allied separations, Green chemistry
- cGMP and key Considerations for the Safety and Cleaning in Manufacturing
- Team project presentations and peer feedback